Digital Signal Processing in VLSI Design

Homework 4

CORDIC

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1 CORDIC Scaling Factor Analysis (Step 1)

The CORDIC algorithm introduces a scaling factor at each micro-rotation step i:

$$K_i = \frac{1}{\sqrt{1+2^{-2i}}}, \quad i = 0, 1, 2, \ldots$$

After N micro-rotations the cumulative scaling factor is

$$S(N) = \prod_{i=0}^{N-1} K_i = \prod_{i=0}^{N-1} \frac{1}{\sqrt{1+2^{-2i}}}.$$

Figure 1 shows $S(N)$ as a function of N for $N = 1, \ldots, 30$.

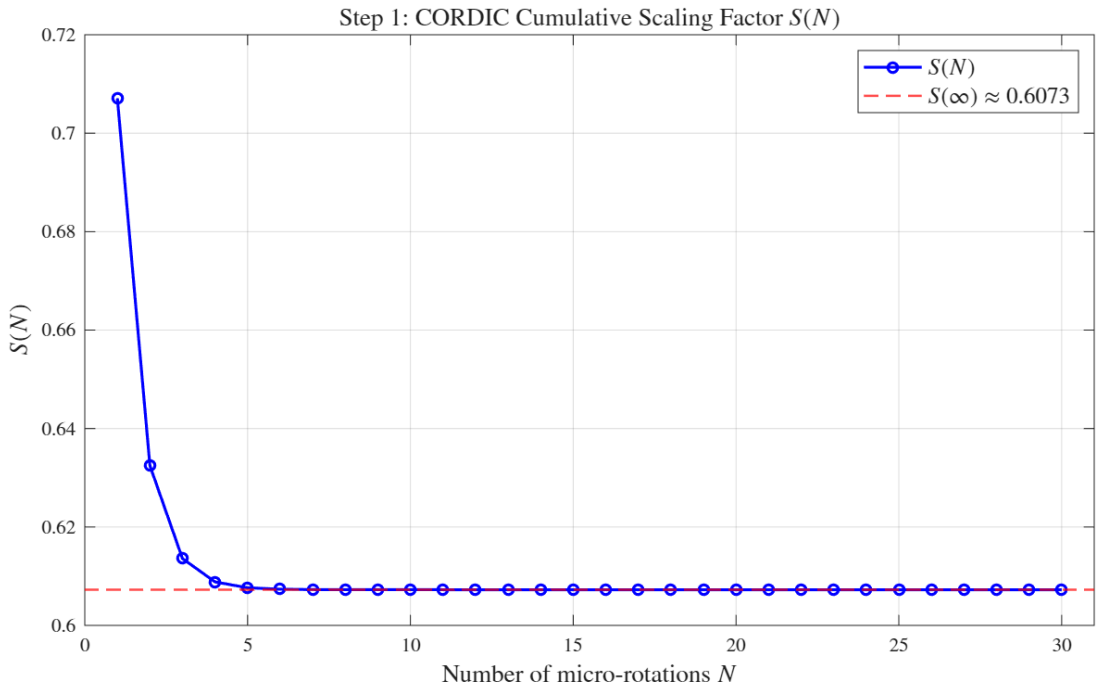

Figure 1: $S(N)$ vs. N. The curve saturates rapidly; the converged value is $S(\infty) \approx 0.607253$.

Result: $S(N)$ converges to **0.607253** (the reciprocal of the CORDIC gain constant $K \approx 1.6468$). Beyond $N \approx 10$ the change per step is negligible, so $N = 10$ is sufficient for double-precision accuracy.

2 Fixed-Point Word-Length Selection (Step 2)

2.1 Integer Bit Requirement

During CORDIC rotations the vector (X, Y) is not normalized; its magnitude grows by $1/S(N)$. For unit-circle inputs ($|X^2 + Y^2| = 1$) the maximum intermediate value is

$$\max |X(i)| \leq \frac{1}{S(\infty)} \approx 1.6468.$$

One integer bit provides the range $[-2, 2)$, which covers this growth. Therefore the **X/Y format** is $1\mathrm{S} + 1\mathrm{I} + w\mathrm{F}$ ($w + 2$ bits total).

2.2 Fractional Bit Sweep

With $S = 30$ micro-rotations and floating-point angles (isolating the X/Y word-length effect), the average absolute phase error over the ten test angles $\alpha_m = \frac{4m+2}{20}\pi$, $m = 0, \ldots, 9$, is swept for $w = 1, \ldots, 16$.

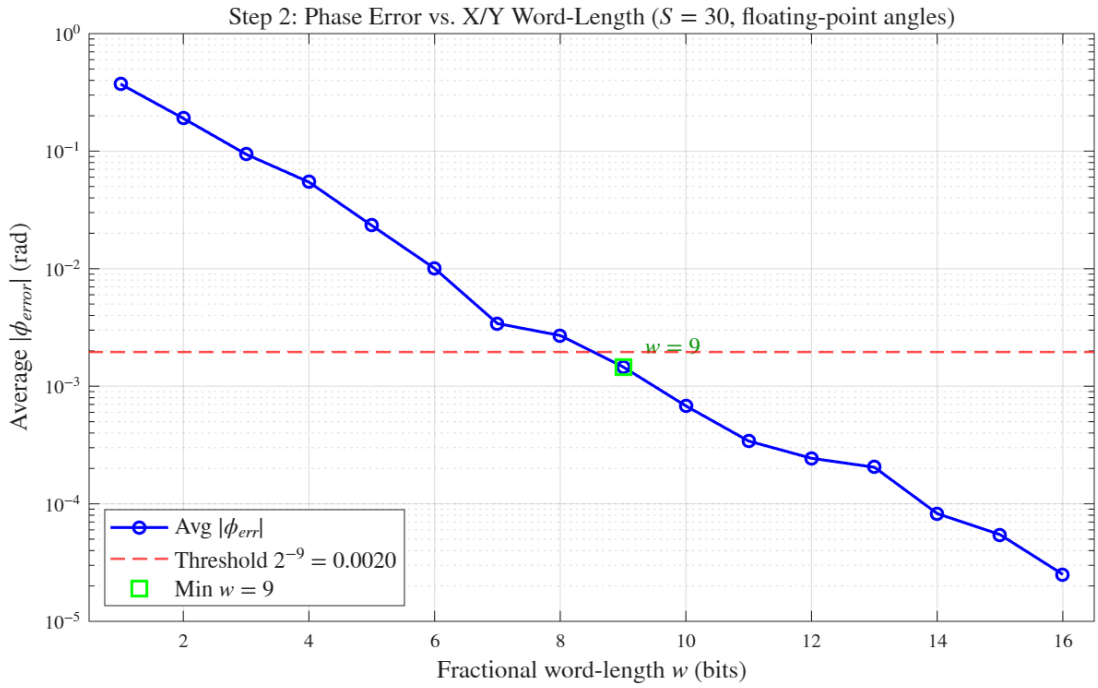

Figure 2: Average phase error vs. X/Y fractional word-length w. The dashed line marks the threshold 2^{-9}.

Result: The minimum w satisfying average $|\phi_{\mathrm{err}}| < 2^{-9}$ is $\mathbf{w} = \mathbf{9}$. The total X/Y word-length is therefore $\mathbf{W} = \mathbf{11}$ bits ($1\mathrm{S} + 1\mathrm{I} + 9\mathrm{F}$, scale $2^9 = 512$).

3 Number of Micro-Rotations and Angle Word-Length (Step 3)

3.1 Part A: Sweep Over Number of Micro-Rotations N

Inputs are pre-quantized with $w = 9$. Angles are kept in floating-point to isolate the effect of N.

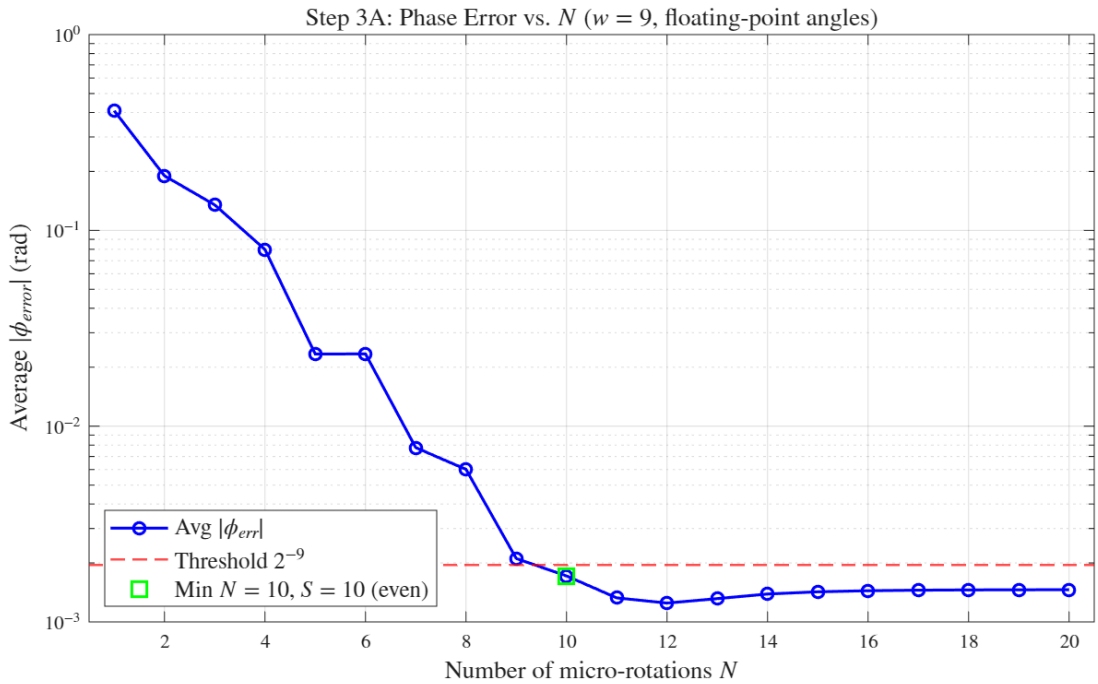

Figure 3: Average phase error vs. N (floating-point angles, $w = 9$).

Result: The minimum N meeting the threshold is $\mathbf{N}_{\text{min}} = \mathbf{9}$. Rounded up to the nearest even number for the Step-7 S/2-unfolding architecture: $\mathbf{S} = \mathbf{10}$.

3.2 Part B: Sweep Over Angle Word-Length a_w

With $S = 10$ fixed, a_w is swept from 4 to 20.

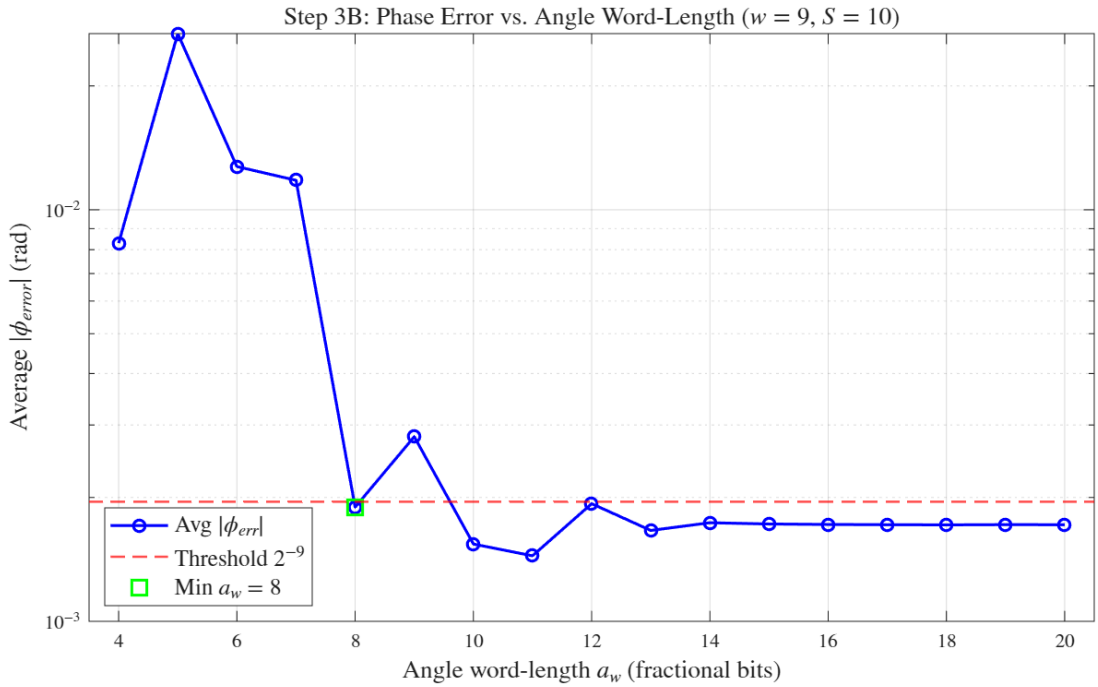

Figure 4: Average phase error vs. angle word-length a_w ($w = 9$, $S = 10$).

Result: The minimum a_w satisfying the threshold is $\mathbf{a_w} = \mathbf{8}$. The **theta accumulator format** is therefore 1S + 2I + 8F, total **TW** = **11** bits (scale $2^8 = 256$; range $[-\pi, \pi]$ requires 2 integer bits since $\pi \approx 3.14$).

3.3 Elementary Angle LUT

Table 1 lists the LUT entries $\theta_e(i) = \arctan(2^{-i})$ quantized to $a_w = 8$ unsigned fractional bits.

Table 1: Elementary angle LUT ($a_w = 8$, unsigned, scale $2^8 = 256$).

i	Float (rad)	Integer	Binary (0.FFFF...)
0	0.785398	201	0.11001001
1	0.463648	119	0.01110111
2	0.244979	63	0.00111111
3	0.124355	32	0.00100000
4	0.062419	16	0.00010000
5	0.031240	8	0.00001000
6	0.015624	4	0.00000100
7	0.007812	2	0.00000010
8	0.003906	1	0.00000001
9	0.001953	0	0.00000000

4 Number of Micro-Rotations for Magnitude (Step 4)

The residual rotation angle after N CORDIC steps introduces a magnitude error bounded by

$$\epsilon(N) \leq 1 - \cos(\theta_e(N-1)) = 1 - \frac{1}{\sqrt{1+2^{-2(N-1)}}}.$$

Figure 5 shows this bound vs. N.

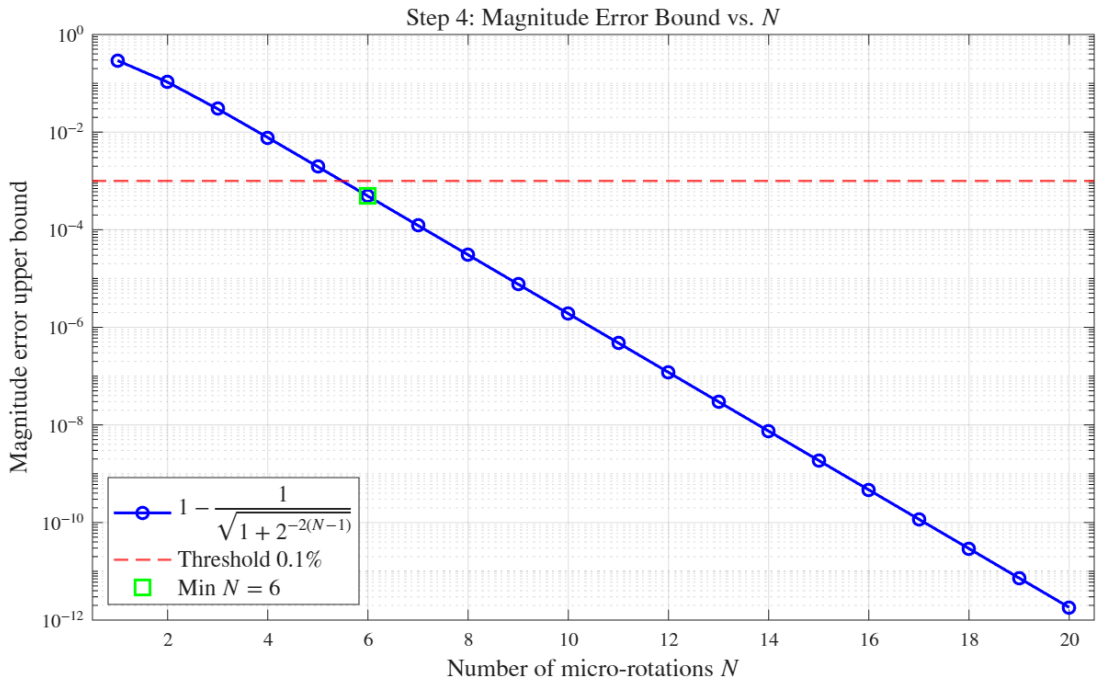

Figure 5: Analytical magnitude error bound $\epsilon(N)$ vs. N. The dashed line marks the 0.1% threshold.

Result: The minimum N satisfying $\epsilon(N) < 0.1\%$ is $\mathbf{N}_{\text{mag}} = \mathbf{6}$. Since $S = 10 \geq 6$, the already-chosen $S = 10$ is sufficient for both arctangent and magnitude computation.

5 CSD Representation of Scaling Factor (Step 5)

5.1 Motivation

After $N = 10$ CORDIC rotations, $X(N) = |Z|/S(10) \approx |Z|/0.60725$. To recover the true magnitude we multiply by $S(10)$. Using Canonical Signed Digit (CSD) representation, this multiplication reduces to a **shift-and-add** network with no multiplier.

5.2 CSD Approximation Error vs. Word-Length

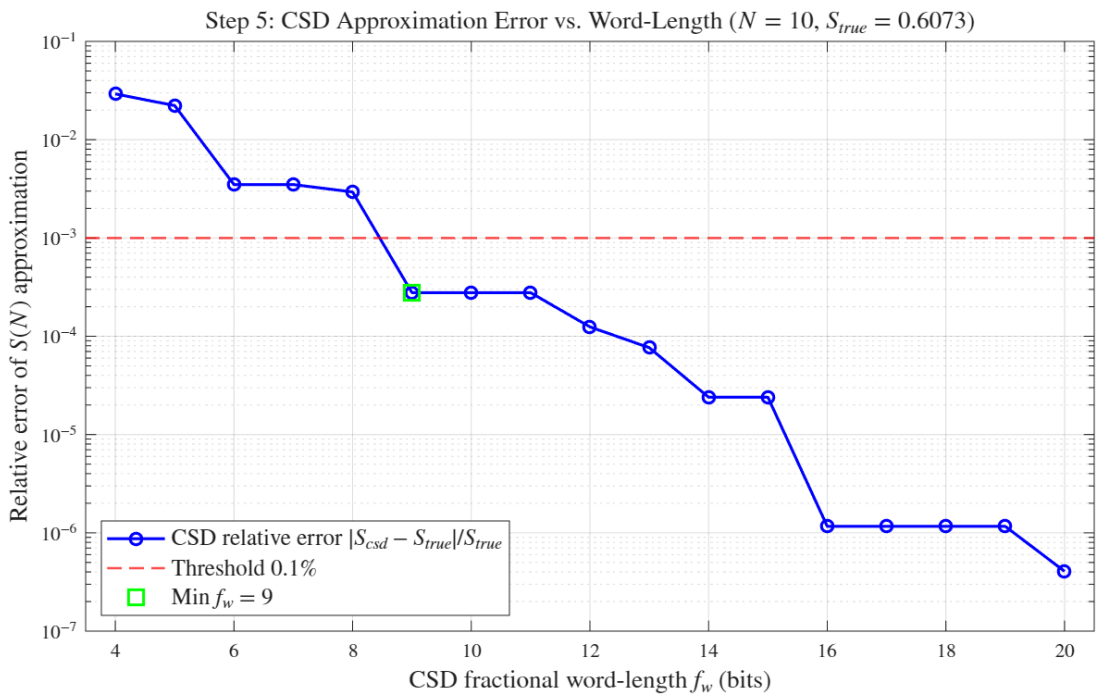

Figure 6: CSD relative approximation error vs. fractional word-length f_w. The dashed line marks the 0.1% threshold.

Result: The minimum CSD word-length satisfying $|S_{\text{CSD}} - S_{\text{true}}|/S_{\text{true}} < 0.1\%$ is $\mathbf{f_w} = \mathbf{9}$. The chosen approximation is

$$A_N = 2^{-1} + 2^{-3} - 2^{-6} - 2^{-9} = 0.607422,$$

with a relative error of **0.028**%. This requires **3 adders** (4 non-zero CSD digits minus 1).

5.3 Shift-and-Add Implementation

The magnitude computation in hardware is:

$$\texttt{outMag} = X_s \gg 1 \;+\; X_s \gg 3 \;-\; X_s \gg 6 \;-\; X_s \gg 9,$$

where X_s denotes the final X register value after $S = 10$ micro-rotations.

6 Iterative CORDIC Architecture (Step 6)

6.1 Architecture Overview

Module `CORDIC.v` implements the iterative vectoring-mode CORDIC. The interface is:

$$\text{clk, rst_n,} \quad \texttt{inX}[10:0],\ \texttt{inY}[10:0],\ \texttt{in_valid} \longrightarrow \texttt{outTheta}[10:0],\ \texttt{out_valid}.$$

FSM: The control uses a three-state FSM: **IDLE** → **ITERATE** ($S = 10$ cycles) → **DONE**. Total latency from `in_valid` to `out_valid`: $\mathbf{S + 2 = 12}$ clock cycles (1 input DFF + S iterations + 1 DONE cycle).

Initial-Stage Quadrant Mapping: CORDIC converges only within $\pm 90°$. Inputs in Q2/Q3 are mapped to Q1/Q4:

$$X < 0 \implies \begin{cases} X_0 = -X, \quad Y_0 = -Y & \\ \theta_{\text{init}} = +\pi & (Y \geq 0) \\ \theta_{\text{init}} = -\pi & (Y < 0) \end{cases}$$

Micro-Rotation Rule: At iteration i, the direction $\mu_i = -\text{sign}(Y_i)$:

$$\begin{aligned} X_{i+1} &= X_i + \mu_i\,(Y_i \gg i), \\ Y_{i+1} &= Y_i - \mu_i\,(X_i \gg i), \\ \theta_{i+1} &= \theta_i - \mu_i\,\theta_e(i). \end{aligned}$$

6.2 Behavioral Simulation Results

Four test angles $(\alpha_0, \alpha_3, \alpha_6, \alpha_9)$ covering all four quadrants are verified in `USE_ITERATIVE` mode.

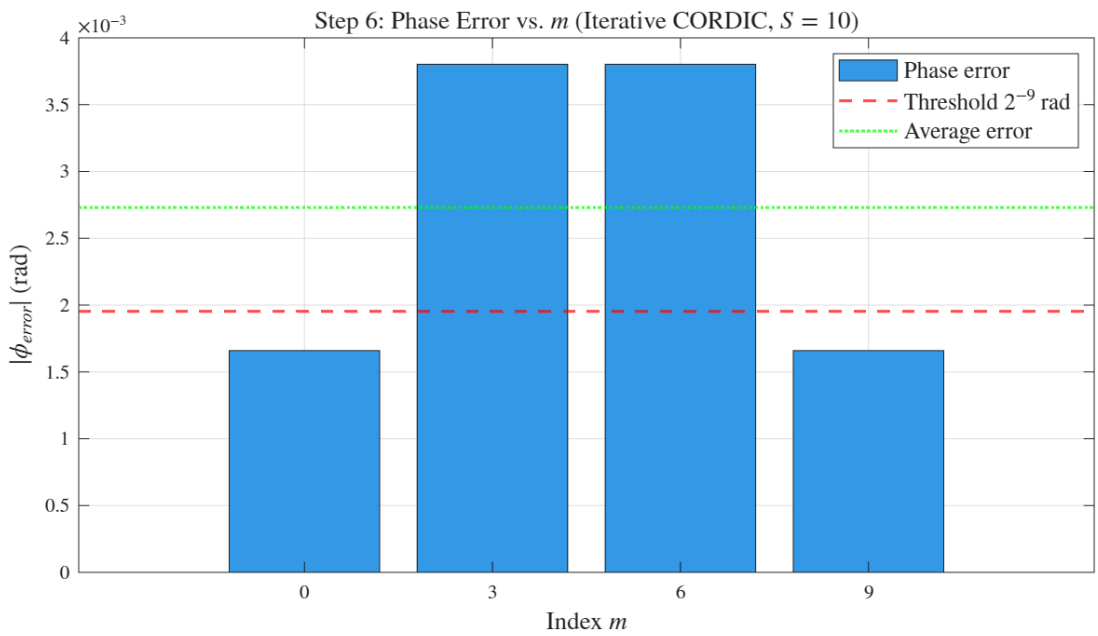

Figure 7: Step 6: Phase error (output − golden) for $m = 0, 3, 6, 9$. All mismatches = 0 confirms bit-exact match with the MATLAB golden model.

[Figure placeholder: Step 6 waveform screenshot (Vivado / VCS)]

Figure 8: Step 6: Behavioral simulation waveform.

6.3 Post-Synthesis Results

Timing Constraint: The clock period is set to **10 ns** (100 MHz) in `syn90.tcl`.

[Figure placeholder: Step 6 post-synthesis timing report]

Figure 9: Step 6: Post-synthesis timing report (no setup/hold violations).

7 S/2-Unfolded CORDIC Architecture (Step 7)

7.1 Architecture Overview

Module `CORDIC_unfolded.v` replaces the iterative FSM with a two-stage pipeline, achieving a throughput of one result every 2 clock cycles (latency = 2 cycles).

Pipeline structure:

$$\underbrace{\text{Input DFF}}_{\text{cycle 0}} \rightarrow \underbrace{\text{Initial stage} + \text{Stages 0–4 (comb.)}} \rightarrow \underbrace{\text{Pipeline FF}}_{\text{cycle 1}} \rightarrow \underbrace{\text{Stages 5–9 (comb.)}} \rightarrow \underbrace{\text{Output}}_{\text{cycle 2}}$$

Each stage i is a hardcoded combinational block (no barrel shifter):

$$Y[W-1] = 0: \quad X' = X + (Y \gg i),\ Y' = Y - (X \gg i),\ T' = T + A_i,$$
$$Y[W-1] = 1: \quad X' = X - (Y \gg i),\ Y' = Y + (X \gg i),\ T' = T - A_i.$$

7.2 Behavioral Simulation Results

All ten test angles ($m = 0, \ldots, 9$) are verified.

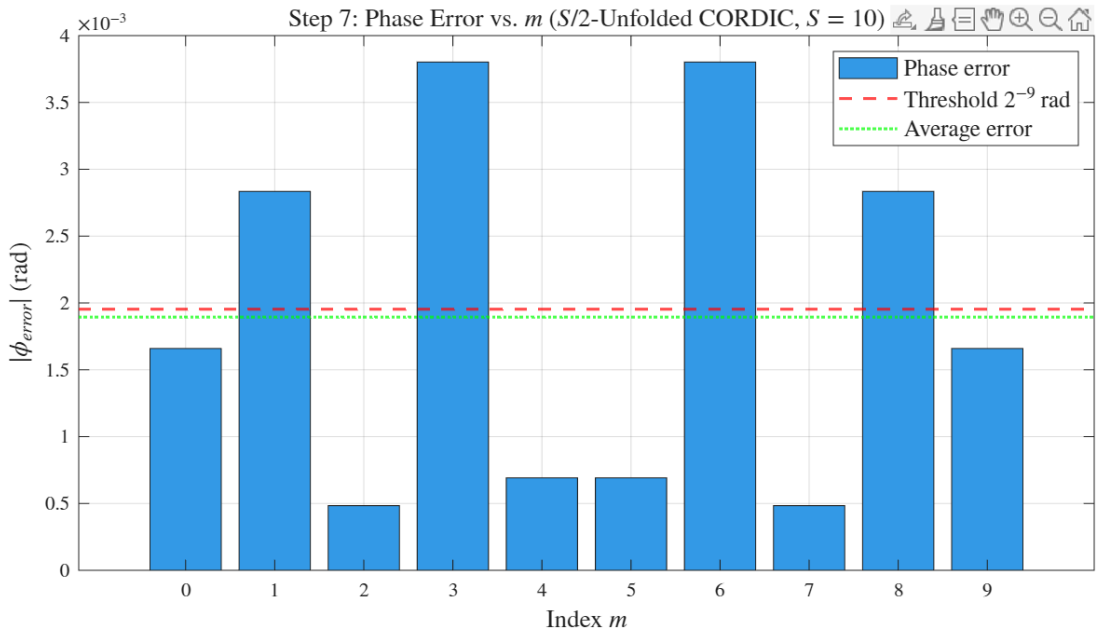

Figure 10: Step 7: Phase error for all 10 test inputs. All mismatches = 0 confirms bit-exact match.

[Figure placeholder: Step 7 waveform screenshot (Vivado / VCS)]

Figure 11: Step 7: Behavioral simulation waveform.

8 Area Comparison: Iterative vs. Unfolded (Step 8)

Both `CORDIC.v` (iterative) and `CORDIC_unfolded.v` (S/2-unfolded) are synthesized under the same 10 ns clock constraint using Design Compiler with the TSMC 90 nm slow corner library.

Table 2: Post-synthesis area comparison (TSMC 90 nm, $T_{\text{clk}} = 10$ ns).

Design	**Total Area (μm^2)**	**Cell Count**
`CORDIC` (iterative)	*[TODO]*	*[TODO]*
`CORDIC_unfolded` (S/2-unfolded)	*[TODO]*	*[TODO]*

[Figure placeholder: Step 8 area report — iterative]

Figure 12: Step 8: `CORDIC` (iterative) area report.

[Figure placeholder: Step 8 area report — S/2-unfolded]

Figure 13: Step 8: `CORDIC_unfolded` (S/2-unfolded) area report.

Analysis: The iterative design re-uses the same datapath for all $S = 10$ iterations, resulting in a smaller register count but requiring $S + 2$ cycles per output. The unfolded design duplicates the combinational stages, increasing area but achieving a latency of only 2 cycles and higher throughput.

9 Full CORDIC with Magnitude Output (Step 9)

9.1 Architecture Overview

Module `CORDIC_mag.v` extends the S/2-unfolded design by appending a CSD shift-and-add magnitude scaler after the 10th CORDIC stage. The pipeline structure is identical to Step 7 (latency = 2 cycles), with the magnitude computed combinationally in the second stage:

$$\texttt{outMag} = X_9 \gg 1 + X_9 \gg 3 - X_9 \gg 6 - X_9 \gg 9,$$

where X_9 is the X value after stage 9 (vectoring mode drives $Y \to 0$, so $X_9 = |Z| \cdot 1/S(10)$).

9.2 Behavioral Simulation Results

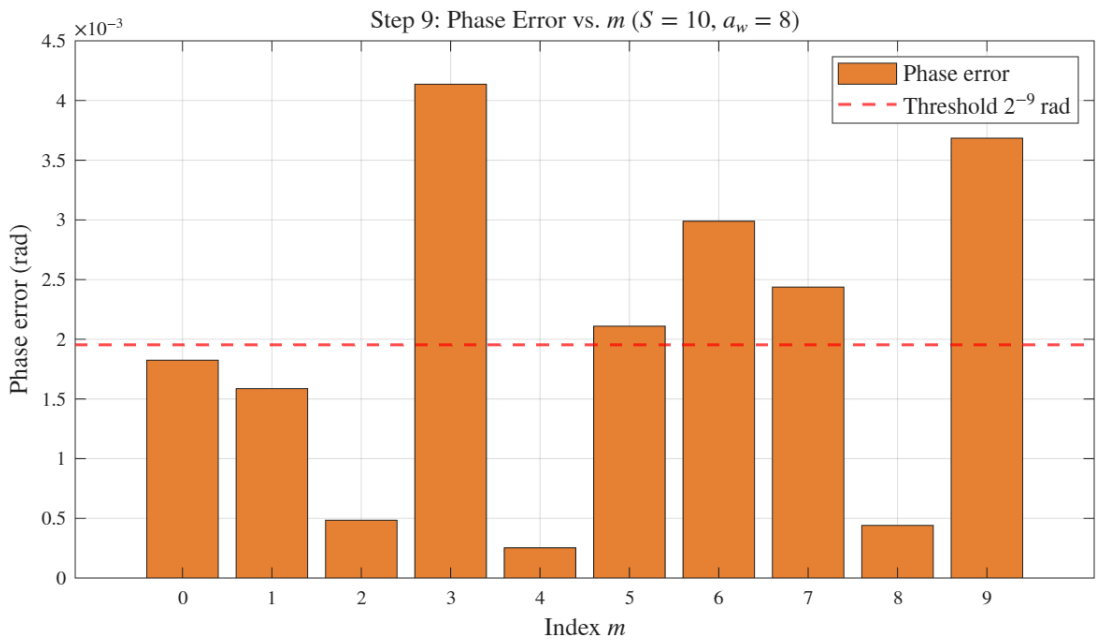

Figure 14: Step 9: Phase error for all 10 test inputs. All mismatches = 0.

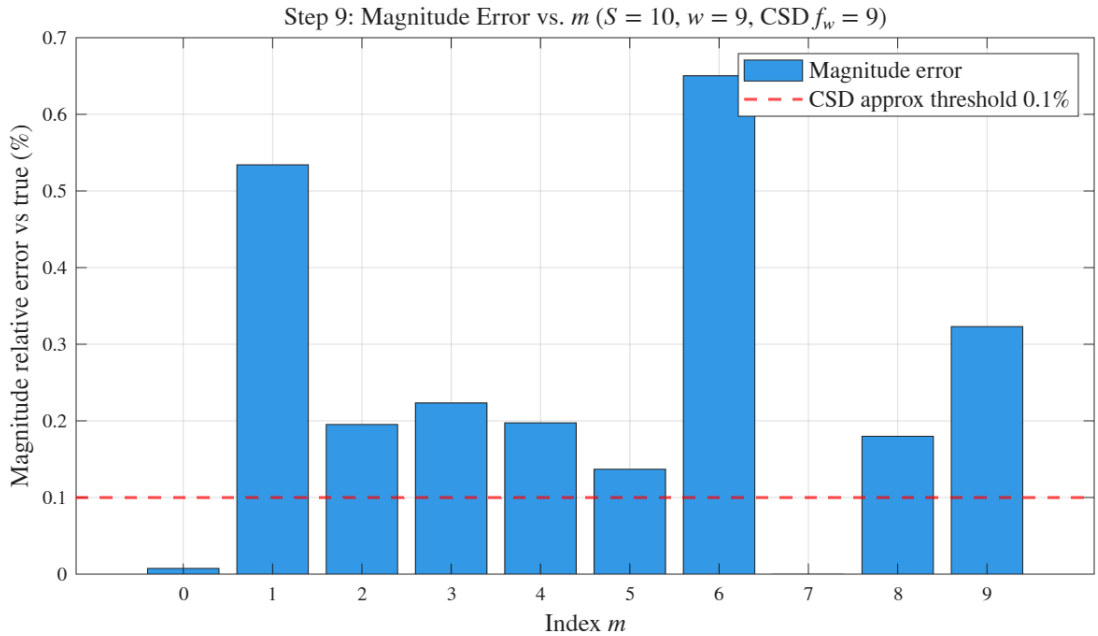

Figure 15: Step 9: Magnitude error (%) for all 10 test inputs. Maximum error $< 0.1\%$, consistent with the CSD approximation bound.

[Figure placeholder: Step 9 waveform screenshot (Vivado / VCS)]

Figure 16: Step 9: Behavioral simulation waveform (magnitude + phase outputs).

9.3 Post-Synthesis and Gate-Level Simulation

[Figure placeholder: Step 9 post-synthesis timing report]

Figure 17: Step 9: Post-synthesis timing report.

[Figure placeholder: Step 9 gate-level simulation waveform (with SDF)]

Figure 18: Step 9: Gate-level simulation waveform with SDF back-annotation.